

INSIGHTS

This project consists of three dashboards – Business Performance Overview, Product Analysis, and Geography Analysis. Each dashboard serves a specific purpose in answer key business questions. The details of each dashboard are as follows:

1. Overall Business Performance

This dashboard contains four **KPIs** – *Total Revenue, Total Cost, Total Profit and %Profit Margin*.

Chart 1 serves as our base graph. It illustrates Total Revenue and Total Profit from 2011-2013. It shows that revenue has decreased every year from **\$35M** to **\$19M**. At the same time, the business has incurred major losses in all three years, however the losses are reducing with time. For instance, the loss incurs in **2011** was **-\$48M** in **2011** which decreased up to **-\$16M** in 2013. This indicates that although the company is operating at a loss, it is moving toward a breakeven point if the progress continues. This insight helped us to select Total Profit as a strong KPI for our further analysis. All subsequent analysis is based on this metric, and the business questions related to profit and loss are explored through both the Product and Geography dashboards.

For a deeper understanding of these KPIs, additional charts present the time-based trends.

Chart 2 shows the trend analysis over the years, indicating that loss decreased from June to Oct. To further investigate efficiency during this period, we analyzed **%Profit Margin** in **Chart 3** to determine whether the business was operating more effectively while losses were reducing. The graph shows that between June to August, **%Profit Margin** reduced to **-126%** which was **-154%** in Jun and Aug.

Since total cost could be our major contributing factor to understand the reason of losses, **Chart 4** analyzes cost trends. It shows the cost decreased from **\$20M** to **\$2M** in 2011, **\$10.3M** to **\$2.4M** in 2012 and **\$7.3M** to **\$2.3M** in 2013. This represents the major contributed factor that helped the business to decreases loss during the period it decreases loss.

2. Product Analysis

We used a drill-down approach to analyze product categories, subcategories, product names, and product classes that incurred the highest losses. This analysis help business identifies the worst performing products through-out from **2011-2013**.

Chart 1 presents total profit and %profit margin across product categories. The analysis shows all product categories incurred losses with the highest loss of **\$25M** coming from **Cell Phone** category. Therefore, we further analyzed selected the Cell Phone category by subcategories in **Chart 2**.

This chart shows that **“Home & Office Phones”** incurred highest losses of \$9.47M while **“Smart phones & PDAs”** and **“Touch screen phones”** generated losses of **\$8.83M** & **\$6.77M**, respectively.

In **Chart 3**, the analysis is further drilled down to 5 worst-performing products with highest losses. The product **“Contoso Phone with 13-Number Memory (210) M301 Grey”** labeled as **“P-1395”**, incurred the maximum loss of **\$0.36M**.

To further analyze the KPIs across the products dimensions, **Chart 4** illustrates the Total Revenue and Total Cost across the 5 worst-performing products in **Cell phones** and **Home & Office Phones**

categories. This analysis indicates that all products have significantly high costs which directly impact profitability.

Finally, **Chart 5** represents the Profits by Product Class of 5 worst-performing products (that are in Chart 4). This shows **Regular Class** products contributes to highest losses of up to **\$1.37M**, while **Deluxe Class** contributed losses of **\$0.35M**.

3. Geography Analysis

For this dashboard, a drill-down approach is used to address the business question.

Chart 1 analyze total revenue and total profit trends across continents. **North America** shows the highest loss, amounting to approximately **-\$69M**, despite generating a total revenue of **\$55M**. Based on this insight, the analysis is further drilled down into countries within this continent.

Chart 2 presents the two countries in North America: **Canada** and **the United States**. The United States accounts for the majority of the loss, approximately **-\$65.4M**, while Canada contributes a smaller loss of around **-\$4M**. To investigate further, revenue and cost trends are analyzed specifically for the United States.

Chart 3 illustrates total revenue and total cost trends in the United States from **2011 to 2013**. The chart indicates a declining trend in both revenue and cost over time, reflecting a consistent downward business performance in this region.

Chart 4 shows that the profit margin percentage is also decreasing over the same period, confirming worsening profitability in the US market.

For deeper analysis, the focus is shifted to individual stores in the US market. **Chart 5** identifies the worst-performing stores in terms of losses: "Contoso Hartford Store" (**-\$0.49M**), "Contoso Westminster Store" (**-\$0.47M**), and "Contoso Leominster Store" (**-\$0.46M**).

Chart 6 presents yearly loss trends for these stores, showing that while losses still exist, they are gradually decreasing over time.

Conclusion

1. Overall Business Performance

Overall, the dashboard analysis indicates that although the company is currently operating at a loss, both declining costs and improving %profit margins suggest a positive trend toward financial recovery and potential breakeven in the future.

2. Product Analysis

The products that generated highest losses includes **Cell Phones** category (**\$25M**), **Home & Office Phones** subcategory (**\$9.47M**), and the following five products:

- Contoso Phone with 13-Number Memory (210) M301 Grey
- Contoso Behind Centrex X15 White
- Contoso 4 Handset Cordless Phone System M86 Black
- Contoso Hybrid system M60 Black
- Contoso multi-line phones M30 White

The analysis suggests that high total cost is the major factor influencing losses. Hence, reducing operational and product related costs could potentially help business improve profitability.

3. Geography Analysis

The analysis reveals that North America, particularly the United States, is the primary driver of overall business losses. Although revenue is present, it is outweighed by significantly higher costs, resulting in consistent negative profit.

At the country level, the United States is responsible for the majority of losses, with declining revenue and cost trends indicating a shrinking but still unprofitable market.

Overall, the business is experiencing a systemic profitability issue in the US market, and improvement efforts should focus on cost reduction, store-level performance optimization, and market restructuring in North America.